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Jongho Kim

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Cornell Johnson Graduate School of Management https://jonghkim.github.io

Education

Cornell University, SC Johnson Graduate School of Management Aug 2021–Present
Ph.D. in Quantitative Marketing
Minor Field: Applied Statistics
Committee: Vrinda Kadiyali (Chair), Emaad Manzoor, Omid Rafieian, Dan Kowal (Statistics)

Korea Advanced Institute of Science and Technology Feb 2015–Feb 2017
M.S. in Management Engineering, Information Systems

Ulsan National Institute of Science and Technology Mar 2010–Aug 2014
B.S. in Computer Science and Engineering, Minor in Finance and Accounting

Journal Publications

Park, J. and **Kim, J.**, 2022. "A Data-Driven Exploration of the Race between Human Labor and Machines in the 21st Century," *Communications of the ACM* [Manuscript] [SSRN] [Website] [Video] [Code]

Choi, S., Kim, W. and **Kim, J.**, 2016. "Compression of Hamiltonian Matrix: Application to Spin-1/2 Heisenberg Square Lattice," *AIP Advances* (**Editor's Pick**) [Manuscript]

Working Papers

Kim, J. and Kadiyali, V., "The Impact of San Diego's Algorithmic Pricing Ban: Evidence from Online Reviews Using Large Language Models" **(Job Market Paper)**

Kim, J., Kowal, D., and Ruppert, D., "Statistical Inference with Large Language Models: A Bayesian Framework for Measurement Errors"

Kim, J., Anand, P., and Kadiyali, V., "The Effects of Rent Control on Renter Experience and Consumption"

Selected Conference and Seminar Presentations

Kim, J., Kowal, D., and Ruppert, D., 2026, "Statistical Inference with Large Language Models: A Bayesian Framework for Measurement Errors," *IO / Empirical Modeling Brown Bag*, Cornell University, Ithaca, NY, USA

Kim, J., Anand, P., and Kadiyali, V., 2025, "The Effects of Rent Control on Consumer Reviews on Rental Housing," *ISMS Marketing Science Conference*, Washington, D.C., USA

Park, J. and **Kim, J.**, 2018. "Leveraging Machine Learning to Reduce Racial Bias on Online Platforms: A Neural Machine Translation Approach," *INFORMS Conference on Information Systems and Technology (CIST)*, Phoenix, Arizona, USA

Kim, J., Cho, D. and Lee, B., 2016. "The Mind Behind Crowdfunding: An Empirical Study of Speech Emotion in Fundraising Success," *International Conference on Information Systems (ICIS)*, Dublin, Ireland (**Most Innovative Research-in-Progress Paper Runner-up**) [Manuscript] [Poster]

Park, J., **Kim, J.** and Lee, B., 2016. "Which Tasks Will Technology Take? A New Systematic Methodology to Measure Task Automation," *International Conference on Information Systems (ICIS)*, Dublin, Ireland [Manuscript] [Poster]

Professional Experience

December & Company Asset Management, Seoul, South Korea Apr 2019–Jul 2021
Senior Researcher / Quantitative Portfolio Manager, Portfolio Research Team

- Develop Generative Models for Textual Factors to Predict and Explain Asset Returns based on SEC-10X (10-K, 10-Q, 10-KSB and 10-QSB)
- Implement a Sector Rotation Strategy through Cross-Sectional Variation over Data-driven Economic Cycles
- Construct Alternative Investment Strategies, especially for Commodity, REITs, Infrastructure and High Dividend-Paying ETFs
- Work as a Substitute for Mandatory Military Service

NICE Pricing and Information, Seoul, South Korea Aug 2016–Apr 2019
Quantitative Researcher, Financial Engineering Center

- Develop Non-Homogeneous Continuous-Time Markov Chain Models (NHCTMC) for Forward Looking Probability of Default under IFRS9 based on Sparse Transition Matrix of Credit Ratings
- Estimate Maximum Likelihood Proxy Portfolio of Fixed-Income Portfolio via Return based Style Analysis (RBSA)
- Implement Implied Volatility Surface of Index Futures Options via Heston Stochastic Volatility Model
- Work as a Substitute for Mandatory Military Service

Digital Economics and Enterprise Research Lab, KAIST, Seoul, South Korea May 2015–Feb 2017

- Empirical Research in Information Systems using Machine Learning and Econometrics (Advisor: Dr. Byungtae Lee)
- Accepted Three Conference Papers in ICIS 2016 and Awarded for the Most Innovative Research-in-Progress Paper Award (Runner-Up) in ICIS 2016
- Government & Industry Projects

  Strategies for Online to Offline (O2O) Transport Services, The Korea Transport Institute (Advisor: Dr. Lee, B.), Sep 2016–Nov 2016

  The Effect of Model-Driven Development (MDD) on Information Systems Maintenance, LG

CNS (Advisor: Dr. Lee, B.), Sep 2015–Dec 2015

Computational Mathematical Science Lab, UNIST, Ulsan, South Korea — Sep 2011–Feb 2015

- Research and Study for Complex Network Theory and Numerical Methods for Differential Equations
- Published a Paper in AIP Advances (Compressed Representation of the Hamiltonian Matrix) and Selected as an Undergraduate Research Opportunity Program (Optimal Control Strategies of Lotka–Volterra Equation)

NewsJAM, Ulsan, South Korea — Jun 2013–Feb 2014

Chief Technology Officer (CTO), Co-Founder

- Developed NLP engines for news analysis including text summarization, key sentence extraction, topic modeling, and text clustering [PyCon Korea 2016]
- Open SW Incubator ($60,000 as award), National IT Industry Promotion Agency, 2013

Honors & Awards

Nagesh Gavirneni Outstanding Doctoral Student Award, Cornell University	2026
AMA-Sheth Doctoral Consortium Fellow	2026
ISMS Doctoral Consortium Fellow	2025
FinTech@Cornell Student Fellowship	2022
Cornell SC Johnson College Doctoral Fellowship	2021–Present
Best Student Paper Award, Post-ICIS 2017 KrAIS Research Workshop	2017
Best Graduate Student Paper Award, KAIST	2017
Most Innovative Research-in-Progress Paper Runner-Up, International Conference on Information Systems (ICIS), Dublin, Ireland	2016
Editor's Picks from the American Institute of Physics, AIP Advances	2016
Scholarship for Academic Excellence among 2015 Entering Class, KAIST	2015
Excellence Prize (Top 1%, 3rd Place over 322 teams) at Hackathon for Big Data, Ministry of Science, ICT and Future Planning	2014
Dean's List for Four Semesters, UNIST	2010, 2012–2014
Undergraduate Research Opportunity Program, Korea Foundation for Advancement of Science & Creativity	2013
National Natural Science and Engineering Scholarship, Korea Student Aid Foundation	2010–2014

Patents

Kim, J., et al., "Automated Trading System and Method," Korean Patent 10-2433324, Assignee: December & Company Inc., 2022. (Application No. 10-2019-0117628, filed Sep 24, 2019)

Teaching

Instructor

Cornell University — Fall 2025

BANA 5440 Introduction to Data Programming, MSBA

StudyPie Oct 2018–Apr 2019

Deep Learning Models for Financial Time Series Prediction [Link]

Guest Lecturer

KAIST FMBA – Introduction to FinTech (Prof. Lee, B.) Mar 2021, Mar 2023, Apr 2024

Investment Strategies in FinTech Industry [Link]

Korea Summer Session on Causal Inference (Prof. Park, J.) Jul 2021

Applications of Machine Learning in Marketing [Link]

KAIST MBA – Business Analytics (Prof. Oh, W.) Apr 2019

Intelligent Video Analytics with Deep Learning [Link]

KAIST IT Management Summer Research Workshop (Prof. Park, J.) Aug 2017

Machine Learning APIs and Web Scraping [Link]

Teaching Assistant

Cornell University

NCCT 5030/5031 Marketing Management Spring, Fall 2023

AEM 4435 / NBA 6390 Data-Driven Marketing Fall 2022

NCC 5030 Marketing Management Summer 2022

Invite-Only Workshops

Econometric Society, Africa Training Workshop in Econometrics, 2024

Services

Ad-hoc Reviewer

International Conference on Information Systems (ICIS), 2021

International Conference on Information Systems (ICIS), 2019

International Conference on Electronic Commerce (ICEC), 2017

Other Information

Work Authorization: U.S. Permanent Residence (I-485 Pending); EAD Expected August 2026; No Sponsorship Required

Language: Korean (Native), English (Fluent)

Programming: Python, C/C++, Java, R, STATA, Matlab, SQL, JavaScript, HTML, CSS

References

Vrinda Kadiyali (Chair)
Nicholas H. Noyes Professor of Management
Professor of Marketing and Economics
SC Johnson Graduate School of Management
Cornell University
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Omid Rafieian
Assistant Professor of Marketing
Demir Sabanci Faculty Fellow
Cornell Tech and SC Johnson College of Business
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Emaad Manzoor
Assistant Professor of Marketing
SC Johnson Graduate School of Management
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Dan Kowal
Associate Professor of Statistics and Data Science
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Selected Research Abstracts

"The Impact of San Diego's Algorithmic Pricing Ban: Evidence from Online Reviews Using Large Language Models," Kim, J. and Kadiyali, V. (Job Market Paper)

In an effort to improve rental housing affordability, San Diego in 2025 prohibited the use of competitors' private data in algorithmic rent pricing. We study its effect on renter satisfaction using 57,262 Google reviews of 1,139 apartments before and after the policy. Each review contains an overall star rating and free text conveying satisfaction with attributes such as rent price and non-price services. When measuring attribute-level satisfaction from review text, the object of interest is often the satisfaction a human reader would understand the review to convey. Yet this satisfaction is not directly observed. Human annotators record it only through coarse ordered categories, from very dissatisfied to very satisfied, and individual ratings are noisy. LLMs can provide such ratings at scale, but introduce a different problem: a rating is generated only after the LLM detects the attribute, and detection may itself depend on the satisfaction conveyed. We develop a Bayesian framework for estimating treatment effects on constructs measured through noisy ordinal human ratings, using large-scale LLM annotations and, in our application, overall ratings as auxiliary information. The framework applies more broadly to constructs commonly assessed using ordered rating scales, such as perceptions and attitudes. Nine months after the policy, overall star ratings decline. Using the framework, we estimate that the policy lowers rent-price satisfaction, with a 60% probability of lower satisfaction under the policy than without it, while raising non-price-service satisfaction, with a 74% probability of higher satisfaction. As the DOJ–RealPage settlement extends a similar restriction nationwide, our findings inform policymakers evaluating the consequences of limiting private-data use in algorithmic pricing.

"The Effects of Rent Control on Renter Experience and Consumption: Evidence from St. Paul's 2022 Ordinance," Kim, J., Anand, P., and Kadiyali, V.

How does housing affordability policy affect consumer experience and grocery spending? We examine St. Paul's 2022 rent control policy using 110,258 renter reviews from 2,285 apartments and grocery sales data from 123 stores in the 18 months after the policy. We compare St. Paul (policy area) and surrounding spillover areas to comparable U.S. cities as controls. Renters rate their apartment experiences higher by 0.474 points (on a 5-point scale) in the policy area and by 0.157 points in the spillover areas. In the 18 months after implementation, no dimension of the multi-attribute service experience worsens, suggesting net satisfaction gains for renters in our time window. Grocery expenditure declines by 8% in the policy area and 3% in the spillover areas over the two years following implementation, through two mechanisms: the exit of higher-income renters from the policy area and likely increased rents in the spillover areas. Renter satisfaction thus improves while grocery demand falls in the affected neighborhoods. The findings inform apartment managers setting service quality under price regulation, grocery retailers adjusting assortment and pricing to changing neighborhood demographics, and policymakers weighing the consequences of rent control for renters, businesses, and spillover areas.

"Statistical Inference with Large Language Models: A Bayesian Framework for Measurement Errors," Kim, J., Kowal, D., and Ruppert, D.

Large language models (LLMs) enable researchers to measure constructs of interest from text at scale. We study treatment effect estimation using LLM-measured outputs as causal outcomes. Standard plug-in estimators can substantially underestimate uncertainty, even when point estimates are unbiased. An LLM's errors are not independent. It misreads semantically similar units in similar ways, inducing correlated errors. This correlation is information, not nuisance, but its structure is unknown a priori. We develop a semiparametric Bayesian framework to estimate treatment effects on latent satisfaction underlying discrete ratings read from text, combining a small labeled sample with large-scale LLM outputs to correct the LLM's measurement errors. Estimating jointly, rather than in stages, allows labeled and unlabeled data to inform each other, while the learned covariance structure induces precision-weighted posterior inference, improving statistical efficiency over methods that assume independent errors. In a randomized experiment, we document that this dependence is real, survives fine-tuning, and escapes conventional clustering, and across simulations our framework maintains valid coverage while improving efficiency. The results show that dependence among an LLM's errors can be exploited rather than ignored.